

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION
CLASS IX • SCIENCE (SUBJECT CODE: 086) • ACADEMIC SESSION 2026-27

SET F — RAPID FIRE SPEED TEST (30 MINS)

CHAPTER 07: WORK, ENERGY & SIMPLE MACHINES

TIME ALLOWED: 30 MINUTES MAXIMUM MARKS: 25 MARKS SERIES: CBS-IX-SCI-SET-F-07

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

GENERAL INSTRUCTIONS (SET F - RAPID FIRE SPEED TEST):

1. The question paper comprises **25 Rapid Fire Multiple Choice Questions** of **1 mark each**.
2. Designed for **Rapid Recall & High-Speed Accuracy**: Recommended target time is **30 Minutes**.
3. All questions are compulsory. There is no negative marking.
4. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

 **SPEED CHALLENGE INSTRUCTIONS:** Answer promptly without looking at notes. Test your instant recall of formulas, scientific definitions, and core NCERT concepts!

Q.1 What is the SI unit of work and energy?

- (A) Watt (B) Joule (J)
(C) Newton (D) Pascal

Q.2 Work done on an object is zero when the angle between force and displacement is:

- (A) 0° (B) 45°
(C) 90° (D) 180°

Q.3 A porter carries a 50 kg luggage bag on his head across a horizontal platform for 20 m. Work done against gravity is:

- (A) 1000 J (B) 9800 J
(C) Zero (D) 500 J

Q.4 What is the kinetic energy of an object of mass m moving with speed v ?

- (A) mv (B) $\frac{1}{2}mv^2$
(C) mgh (D) $\frac{1}{2}m^2 v$

Q.5 What is the gravitational potential energy of an object of mass m raised to height h ?

- (A) mgh (B) $\frac{1}{2}mgh$
(C) $m v^2$ (D) $F \times t$

Q.6 If the speed of a moving body is doubled, its kinetic energy becomes:

- (A) 2 times (B) 4 times
(C) 8 times (D) Half

Q.7 What is the commercial unit of electrical energy commonly called a 'unit' on electricity bills?

- (A) Joule (B) Kilowatt-hour (kWh)
(C) Watt-second (D) Horsepower

Q.8 How many Joules are equivalent to 1 kilowatt-hour (1 kWh)?

- (A) $3.6 \times 10^3 \text{ J}$ (B) $3.6 \times 10^6 \text{ J}$
(C) 1000 J (D) 3600 J
-

Q.9 What is the SI unit of power?

- (A) Joule (B) Watt (W)
(C) Newton (D) Calorie
-

Q.10 How many Watts are in 1 Horsepower (1 hp)?

- (A) 500 W (B) 746 W
(C) 1000 W (D) 3600 W
-

Q.11 When a stone is thrown vertically upward, work done by gravity during ascent is:

- (A) Positive (B) Negative
(C) Zero (D) Infinite
-

Q.12 What happens to the total mechanical energy of an oscillating simple pendulum in a vacuum?

- (A) It increases continuously (B) It remains constant (conserved)
(C) It decreases to zero (D) It oscillates between 0 and infinity
-

Q.13 A compressed spring in a toy gun possesses which form of energy?

- (A) Gravitational potential energy (B) Elastic potential energy
(C) Nuclear energy (D) Kinetic energy
-

Q.14 An electric bulb of power 60 W is lighted for 10 hours. How much electrical energy is consumed?

- (A) 0.6 kWh (B) 6.0 kWh
(C) 600 kWh (D) 0.06 kWh
-

Q.15 A body of mass 2 kg is dropped from a height of 10 m. What is its kinetic energy just before hitting ground ($g = 9.8 \text{ m/s}^2$)?

- (A) 98 J (B) 196 J
(C) 20 J (D) 392 J
-

Q.16 The work-energy theorem states that net work done on a body equals:

- (A) Change in potential energy (B) Change in kinetic energy
(C) Total power (D) Change in momentum
-

Q.17 In a hydroelectric dam, which energy transformation occurs?

- (A) Thermal to electrical (B) Potential energy of water $\rightarrow$ Kinetic energy $\rightarrow$ Electrical energy
(C) Chemical to nuclear (D) Sound to light
-

Q.18 Can a machine have an efficiency (η) greater than 100%?

- (A) Yes, if it has 10 pulleys (B) No, it violates the Law of Conservation of Energy
(C) Only on the Moon (D) Yes, using ball bearings
-

Q.19 Mechanical Advantage (MA) of a simple machine is defined as the ratio of:

- (A) Effort to Load (B) Load to Effort ($\frac{\text{Load}}{\text{Effort}}$)
(C) Distance of effort to distance of load (D) Power input to power output

Q.20 A wheelbarrow is an example of which class of lever?

- (A) Class 1 lever (B) Class 2 lever (Load in middle)
(C) Class 3 lever (D) Compound pulley

Q.21 What is the Velocity Ratio (VR) of a single fixed pulley?

- (A) 1 (B) 2
(C) 0.5 (D) 4

Q.22 When a satellite orbits the Earth in a circular path, work done by Earth's gravity on the satellite is:

- (A) Zero, because gravitational force is perpendicular to tangential motion (B) Positive
(C) Negative (D) Equal to its mass

Q.23 An engine lifts a load of 1000 N through 5 m in 10 s. What is its power output?

- (A) 5000 W (B) 500 W
(C) 50 W (D) 200 W

Q.24 Work done by kinetic friction on a sliding box is always:

- (A) Positive (B) Negative
(C) Zero (D) Undefined

Q.25 What is the relationship between Mechanical Advantage (MA), Velocity Ratio (VR), and Efficiency (η)?

- (A) $\text{MA} = \eta \times \text{VR}$ (B) $\text{VR} = \eta \times \text{MA}$
(C) $\eta = \text{MA} \times \text{VR}$ (D) $\text{MA} = \text{VR} / \eta$

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET F)

01 (A) (B) (C) (D)	02 (A) (B) (C) (D)	03 (A) (B) (C) (D)	04 (A) (B) (C) (D)	05 (A) (B) (C) (D)
06 (A) (B) (C) (D)	07 (A) (B) (C) (D)	08 (A) (B) (C) (D)	09 (A) (B) (C) (D)	10 (A) (B) (C) (D)
11 (A) (B) (C) (D)	12 (A) (B) (C) (D)	13 (A) (B) (C) (D)	14 (A) (B) (C) (D)	15 (A) (B) (C) (D)
16 (A) (B) (C) (D)	17 (A) (B) (C) (D)	18 (A) (B) (C) (D)	19 (A) (B) (C) (D)	20 (A) (B) (C) (D)
21 (A) (B) (C) (D)	22 (A) (B) (C) (D)	23 (A) (B) (C) (D)	24 (A) (B) (C) (D)	25 (A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & RAPID EXPLANATIONS (SET F)

Q. No.	Correct	Scientific Rationale & Formula Summary
Q.1	(B)	The Joule (J) is the SI unit of both work and energy ($1\text{ J} = 1\text{ N} \cdot \text{m}$).
Q.2	(C)	$W = F \cos \theta$. When $\theta = 90^\circ$, $\cos 90^\circ = 0$, so work done is zero.
Q.3	(C)	Gravity acts vertically downward while displacement is horizontal ($\theta = 90^\circ$; $W = 0$).
Q.4	(B)	Kinetic energy is given by $E_k = \frac{1}{2}mv^2$.
Q.5	(A)	$E_p = mgh$, where g is the acceleration due to gravity.
Q.6	(B)	$E_k \propto v^2$. If speed doubles ($2v$), kinetic energy quadruples ($2^2 = 4$ times).
Q.7	(B)	$1\text{ commercial unit} = 1\text{ kilowatt-hour (kWh)} = 3.6 \times 10^6\text{ J}$.
Q.8	(B)	$1\text{ kWh} = 1000\text{ W} \times 3600\text{ s} = 3.6 \times 10^6\text{ J}$ (3.6 MegaJoules).
Q.9	(B)	Power is the rate of doing work ($P = W/t$). Its SI unit is the Watt ($1\text{ W} = 1\text{ J/s}$).
Q.10	(B)	One standard imperial mechanical horsepower equals exactly 746 Watts.
Q.11	(B)	Displacement is upward while gravity acts downward ($\theta = 180^\circ$; $\cos 180^\circ = -1$; $W < 0$).
Q.12	(B)	In the absence of dissipative friction, total mechanical energy ($E_k + E_p$) remains constant.
Q.13	(B)	Work done in deforming the spring against elastic restoring forces is stored as elastic potential energy.
Q.14	(A)	$\text{Energy} = P \times t = 0.060\text{ kW} \times 10\text{ h} = 0.6\text{ kWh}$.
Q.15	(B)	By conservation of energy: $E_k = E_p = mgh = 2 \times 9.8 \times 10 = 196\text{ J}$.
Q.16	(B)	$W_{\text{net}} = \Delta E_k = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$.
Q.17	(B)	Stored water potential energy converts to kinetic energy as it rushes down, which spins turbines generating electricity.
Q.18	(B)	Work output can never exceed work input without violating the law of conservation of energy.
Q.19	(B)	$\text{Mechanical Advantage} = \frac{\text{Load (Resistance)}}{\text{Effort (Applied force)}}$.
Q.20	(B)	In a wheelbarrow, the load is between the fulcrum (wheel axle) and the effort (handles), defining Class 2.
Q.21	(A)	In a single fixed pulley, the effort moves the exact same distance as the load ($VR = 1$).
Q.22	(A)	Centripetal gravitational pull is orthogonal ($\theta = 90^\circ$) to the satellite's instantaneous displacement ($W = 0$).
Q.23	(B)	$P = \frac{W}{t} = \frac{F \times s}{t} = \frac{1000 \times 5}{10} = \frac{5000}{10} = 500\text{ W}$.
Q.24	(B)	Kinetic friction always opposes the direction of sliding displacement ($\theta = 180^\circ$, $W < 0$).
Q.25	(A)	$\text{Efficiency } \eta = \frac{\text{MA}}{\text{VR}} \Rightarrow \text{MA} = \eta \times \text{VR}$.